VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System

MILESTONE 4

Testing, Deployment & Documentation

Week 7 & 8

Milestone Progress Document

1. Project Overview

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System is an AI-powered manufacturing quality-inspection platform intended to detect product defects from images, identify quality issues, classify defect types, and provide production analytics. The platform is designed to reduce manual inspection effort, improve product quality, minimize manufacturing defects, and increase production efficiency.

2. Milestone 4 Objective

Validate the complete VisionInspect AI platform, improve model and inspection performance, deploy the frontend and backend using the planned container/cloud approach, and prepare technical documentation and the final demonstration.

3. Scope of Work

- Validated defect-detection accuracy and classification quality.
- Optimized computer-vision models and inspection performance.
- Improved dashboard responsiveness and reporting quality.
- Prepared Docker-based deployment and cloud-environment deployment using the planned AWS/Azure approach.
- Prepared technical documentation and presentation material.
- Demonstrated the complete VisionInspect AI platform end to end.

4. Planned System Modules

- Model validation
- Platform testing
- Docker/cloud deployment
- Documentation and presentation

5. Key Deliverables

- Model testing and validation results
- Optimized inspection workflow
- Responsive analytics and reporting dashboard
- Deployed frontend and backend
- Technical documentation
- Final presentation and end-to-end demonstration

6. Expected / Recorded Outcomes

- Deployment and testing experience.
- Improved defect-detection accuracy and platform usability.

- Live deployment and final demonstration readiness.
- Professional project documentation and presentation.

7. Technology Stack

Backend: Python (FastAPI) | Frontend: React.js / Next.js | Database: PostgreSQL / MongoDB

AI & Computer Vision: OpenCV, TensorFlow, PyTorch, YOLO, CNN models, Scikit-learn, NumPy and Pandas.

Deployment & Development: Docker, Docker Compose, AWS/Azure, Git/GitHub, VS Code and Postman.

8. Performance & Evaluation Focus

The project specification identifies defect-detection accuracy, precision, recall, F1-score, mAP, inspection automation rate, defect identification accuracy, false-defect detection rate, image-processing speed, inspection response time, and concurrent inspection capability as key metrics.

9. Challenges / Notes

This milestone document is based on the supplied VisionInspect AI project specification. Where implementation evidence, measured model metrics, screenshots, deployment URLs, or issue logs were not included in that source, this document does not invent those details. Actual measured results and screenshots can be added to the GitHub copy before submission.

10. Next Milestone / Continuation

Maintain the deployed platform, monitor model and system performance, and use inspection analytics for future continuous improvement.

Conclusion

Milestone 4 establishes the planned progression of the VisionInspect AI system toward an end-to-end manufacturing defect detection and quality inspection platform. The work in this milestone provides the foundation for the next stage of implementation and final demonstration.